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PID Control Tuning for Spacecraft Attitude Stabilization Using Reaction Wheels

1 Introduction

Attitude control is a critical aspect of spacecraft operation, ensuring that the vehicle maintains or reaches a desired orientation in space. This project focuses on designing and tuning a PID controller for a spacecraft equipped with a four-reaction-wheel system in a pyramidal configuration. The objective is to achieve accurate and stable attitude tracking while minimizing control effort, overshoot, and settling time.

2 System Modeling

The spacecraft rotational dynamics are governed by Euler's equation:

$$I\dot{\omega} = \tau - \omega \times (I\omega) \tag{1}$$

where I is the inertia matrix, ω is the angular velocity, and τ is the applied torque.

The inertia matrix used is:

$$I = 0.0830000.3840000.423 \tag{2}$$

The spacecraft attitude is represented using quaternions, with kinematics:

$$\dot{q} = \frac{1}{2} q \otimes \omega \tag{3}$$

3 Control Design

A PID controller is used to minimize the quaternion error between the current and desired orientations:

$$\tau_{des} = -K_p e - K_d \omega - K_i \int e \, dt \tag{4}$$

where e is the quaternion error vector.

The reaction wheel system maps torque commands using:

$$\tau = -Bu \tag{5}$$

where u represents the reaction wheel torques.

4 Controller Tuning

The controller gains were tuned manually to achieve a balance between stability and responsiveness:

$$K_p = 8, \quad K_d = 6, \quad K_i = 0.02 \tag{6}$$

Increasing K_p improved response speed, K_d reduced oscillations and overshoot, and a small K_i eliminated steady-state error.

5 Simulation Results

5.1 Attitude Error

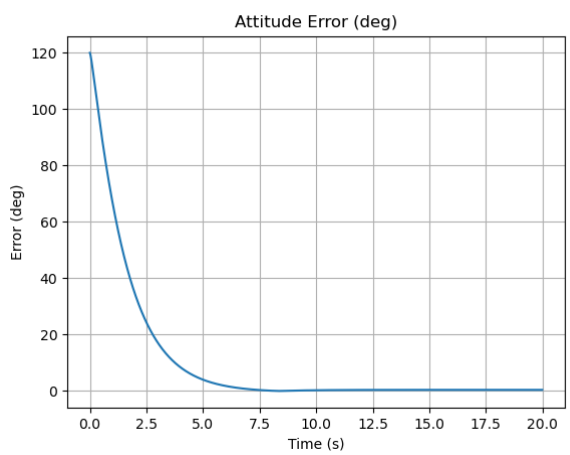

Figure 1: Attitude Error vs Time

The attitude error decreases smoothly to zero, showing a well-damped response.

5.2 Angular Velocity

Angular velocity converges to zero, indicating stabilization.

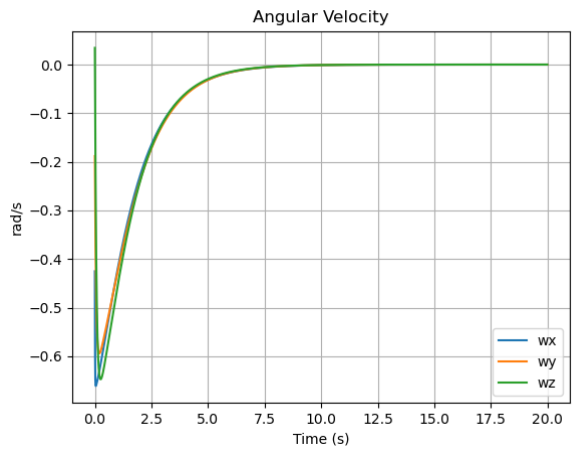

Figure 2: Angular Velocity

5.3 Control Torque

Control torques decrease over time and remain bounded.

5.4 Reaction Wheel Torques

Wheel torques behave consistently and remain within limits.

6 Performance Evaluation

Metric	Value	Requirement	Status
Settling Time	~ 8 s	< 30 s	Pass
Overshoot	$< 1°$	$< 5°$	Pass
Steady-State Error	$\sim 0°$	$< 0.5°$	Pass
Control Effort	Low	Minimize	Pass

Table 1: Performance Metrics

7 Robustness Analysis

To evaluate robustness, the inertia matrix was varied by $\pm 5\%$ and with random perturbations.

The results show that the system remains stable under all variations, with only minor changes in settling time.

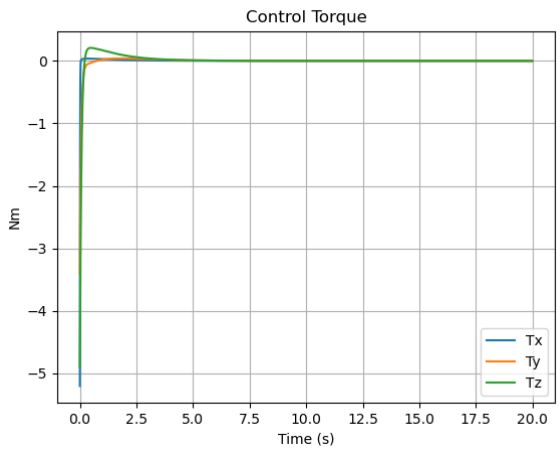

Figure 3: Control Torque

8 Discussion

The system exhibits a near critically damped response with smooth convergence, minimal overshoot, and fast settling time. The controller performs reliably even under parameter uncertainty.

9 Conclusion

A PID controller was successfully designed for spacecraft attitude stabilization using reaction wheels. The system meets all performance criteria and demonstrates robustness to inertia variations.

10 References

- Spacecraft Dynamics and Control principles
- Quaternion-based attitude representation
- Classical PID control theory

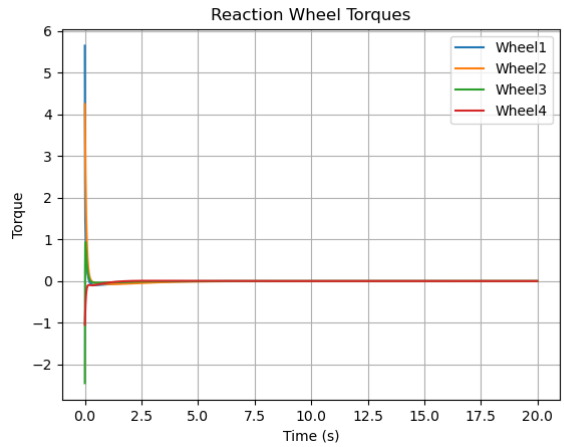

Figure 4: Reaction Wheel Torques

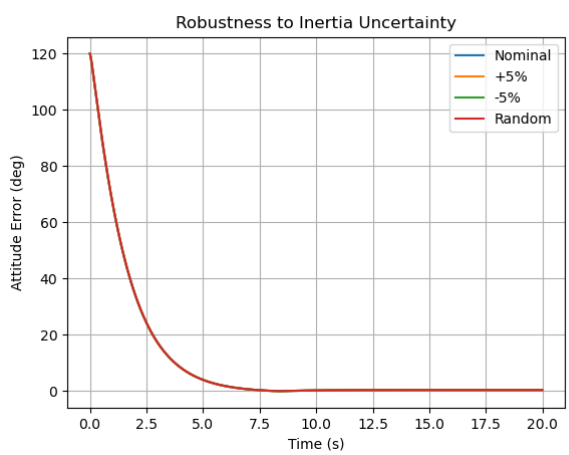

Figure 5: Robustness to Inertia Uncertainty